

Mendelian Randomization Study of B-Type Natriuretic Peptide and Type 2 Diabetes: Evidence of Causal Association from Population Studies

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BACKGROUND Genetic and epidemiological evidence suggests an inverse association between B-type natriuretic peptide (BNP) levels in blood and risk of type 2 diabetes (T2D), but the prospective association of BNP with T2D is uncertain, and it is unclear whether the association is confounded. **METHODS AND FINDINGS** We analysed the association between levels of the N-terminal fragment of pro-BNP (NT-pro-BNP) in blood and risk of incident T2D in a prospective case-cohort study and genotyped the variant rs198389 within the BNP locus in three T2D case-control studies. We combined our results with existing data in a meta-analysis of 11 case-control studies. Using a Mendelian randomization approach, we compared the observed association between rs198389 and T2D to that expected from the NT-pro-BNP level to T2D association and the NT-pro-BNP difference per C allele of rs198389. In participants of our case-cohort study who were free of T2D and cardiovascular disease at baseline, we observed a 21% (95% CI 3%-36%) decreased risk of incident T2D per one standard deviation (SD) higher log-transformed NT-pro-BNP levels in analysis adjusted for age, sex, body mass index, systolic blood pressure, smoking, family history of T2D, history of hypertension, and levels of triglycerides, high-density lipoprotein cholesterol, and low-density lipoprotein cholesterol. The association between rs198389 and T2D observed in case-control studies (odds ratio = 0.94 per C allele, 95% CI 0.91-0.97) was similar to that expected (0.96, 0.93-0.98) based on the pooled estimate for the log-NT-pro-BNP level to T2D association derived from a meta-analysis of our study and published data (hazard ratio = 0.82 per SD, 0.74-0.90) and the difference in NT-pro-BNP levels (0.22 SD, 0.15-0.29) per C allele of rs198389. No significant associations were observed between the rs198389 genotype and potential confounders. **CONCLUSIONS** Our results provide evidence for a potential causal role of the BNP system in the aetiology of T2D. Further studies are needed to investigate the mechanisms underlying this association and possibilities for preventive interventions. Please see later in the article for the Editors' Summary.